Necessity of Memorization for Generalization in Long-Tailed Distributions (Feldman 2020)

Achieving Bayes-optimal generalization on long-tailed data distributions mathematically necessitates memorizing training sample labels from rare sub-populations, creating a fundamental tension with differential privacy.

GRADE: HIGH | STATUS: STABLE | DOMAIN: TECHNICAL | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

When the underlying data-generating distribution is heavy-tailed—comprising a massive frequency tail of rare sub-populations—any learning algorithm that achieves close-to-optimal generalization error must memorize individual labels and specific features of training instances appearing only once or twice in the dataset.

EVIDENCE AND CAUSAL ASSESSMENT

Mathematical proofs demonstrate that in distributions where the frequency of sub-populations decays as a power law, the cumulative probability mass of rare atypical examples dominates total test loss; learning algorithms that suppress atypical instance memorization cannot achieve Bayes-optimal generalization error.

EFFECT SIZE AND PERSISTENCE

Empirical measurements on ImageNet show that memorization of rare examples accounts for substantial accuracy gains (+3.7% accuracy on ImageNet directly attributable to memorized instances), establishing that memorization in deep learning is not merely an over-parameterization bug.

BIAS, HETEROGENEITY, AND CONFOUNDING

Classified as n/a (formal theorem). The mathematical trade-off between memorization and generalization on long-tailed distributions is fundamental, establishing an irreducible structural conflict with strict differential privacy ($\varepsilon << 1$).

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

The theorem does not imply that all memorization is useful; memorizing arbitrary label noise or random data artifacts degrades out-of-distribution performance without conferring inductive benefit.

WHAT ACTION IS REASONABLY JUSTIFIED

Acknowledge the fundamental structural trade-off between strict differential privacy and optimal utility on tail sub-populations; avoid naive claims that zero memorization can be achieved without sacrificing minority-group performance.

VERIFIED SOURCES & DOI IDENTIFIERS

- Feldman, V. (2020). Does Learning Require Memorization? A Short Tale about a Long Tail. Proceedings of the 52nd Annual ACM SIGACT Symposium on Theory of Computing (STOC 2020), 954-959. https://arxiv.org/abs/1906.05271